

Subject Description Form

Subject Code	AMA3640
Subject Title	Statistical Inference
Credit Value	3
Level	3
Pre-requisite	Calculus and Linear Algebra (AMA1007) or Basic Mathematics II – Calculus and Linear Algebra (AMA1120) or Calculus for Engineers (AMA1130) or Calculus (AMA1131) or Foundation Mathematics for Accounting and Finance (AMA1500) or Data Analytics Fundamentals (AMA1611/DSAI1102) or Calculus (AMA1702) and Introduction to Statistics for Business (AMA1501) or Introduction to Statistics (AMA1502/AMA1602) or Probability and Engineering Statistics (AMA2104) or Probability and Distributions (AMA2691) or Principles of Data Science (DSAI1103) or Mathematical Methods for Finance (AMA2703) or equivalent
Exclusion	Statistical Inference (AMA364)
Objectives	This subject is to enable students to understand the theory of statistical inference and apply it to data analysis.
Intended Learning Outcomes	Upon satisfactory completion of the subject, students should be able to: - master the fundamental concepts of point estimation and interval estimation; - apply methods of estimation, criteria of assessing a good estimator to determine the distribution and statistical properties of an estimator; - perform tests of hypotheses relating population parameters and to judge the appropriateness and goodness of tests.
Subject Synopsis/ Indicative Syllabus	<i>Estimation: (17 hours)</i> Statistic, unbiased estimator, consistent estimator. Minimum variance unbiased estimator. Efficiency of an unbiased estimator. Sufficiency, Factorisation theorem. Information matrix. Cramér-Rao lower bound. Relative efficiency. Method of moments. Likelihood, maximum likelihood (ML) estimation. Properties of ML estimators. Iterative solutions of ML estimating equation. <i>Hypothesis testing: (11 hours)</i> Significance test. Types of error, power of test. Neyman-Pearson theorem. Uniformly most powerful test. Generalised likelihood ratio test.

	<i>Bayesian inference: (11 hours)</i> Bayes' formula, Prior and Posterior distributions. Uniform prior, Conjugate prior. Bayes' solution to decision problem. Loss function, Bayesian estimation. Credible interval. Predictive inference.				
Teaching/Learning Methodology	The subject will be delivered mainly through lectures and tutorials. The lectures will be conducted to introduce the statistical inference concepts in the syllabus, which are then reinforced by learning activities involving demonstration, tutorial exercise and assignments/quizzes.				
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
			a	b	c
	1. Assignments	20%	✓	✓	
	2. Quizzes	20%	✓	✓	
	3. Examination	60%	✓	✓	✓
	Total	100 %			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: The subject focuses on knowledge, skill and understanding of <u>Statistical Inference</u> , thus, <u>Exam-based assessment</u> is the most appropriate assessment method, including 60% examination. Moreover, 40% worth of assignments and quizzes are included as a component of continuous assessment so as to keep the students in progress. Continuous Assessment comprises of assignments and/or quizzes. A written examination is held at the end of the semester.				
Student Study Effort Expected	Class contact:				
	• Lecture			26 Hrs.	
	• Tutorial			13 Hrs.	
	Other student study effort:				
	• Assignment			30 Hrs.	
	• Self-study			36 Hrs.	
	Total student study effort			105 Hrs.	
Reading List and References	<u>Textbook:</u> Hogg, R.V., McKean, J.W. & Craig, A.T. Introduction to Mathematical Statistics 7 th edition Prentice Hall 2012 <u>References:</u>				

	Bain, L.J. & Engelhardt, M.	Introduction to Probability and Mathematical Statistics 2 nd edition	Duxbury Press 2000
	Casella, G. & Berger, R.L.	Statistical Inference 2 nd edition	Duxbury Press 2001
	Garthwaite, P., Jolliffe, I. & Byron, J.	Statistical Inference 2 nd edition	Oxford Science Publication 2002
	Mood, A.M.	Introduction to the Theory of Statistics 3 rd edition	McGraw-Hill 1974